Panel Data Analysis

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Example: A Natural Experiment in Education

In a famous paper in the *American Economic Review*, Esther Duflo examines the impacts of a large school construction program in Indonesia

Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment

By Esther Duflo*

Between 1973 and 1978, the Indonesian government engaged in one of the largest school construction programs on record. Combining differences across regions in the number of schools constructed with differences across cohorts induced by the timing of the program suggests that each primary school constructed per 1,000 children led to an average increase of 0.12 to 0.19 years of education, as well as a 1.5 to 2.7 percent increase in wages. This implies estimates of economic returns to education ranging from 6.8 to 10.6 percent. (*JEL* I2, J31, O15, O22)

Example: A Natural Experiment in Education

The Sekolar Dasar INPRES program (1973–1979):

- Oil crisis creates large windfall for Indonesia
- Suharto uses oil money to fund school construction
- Close to 62,000 schools built by national gov't
  - Approximately 1 school built per 500 school-age children
- More schools built in areas which started with fewer schools
- Schools intended to promote equality, national identity

Example: A Natural Experiment in Education

Do children who were born into areas with more newly built INPRES primary schools get more education? Do they earn more as adults?

Strategy: **difference-in-differences estimation**

- Data on children born before and after program (pre vs. post)
  - Children aged 12 and up in 1974 did not benefit from program
  - Children aged 6 and under were young enough to be treated
- Data on children born in communities where many schools were built (treatment), those where few schools were built (comparison)
  - Partition sample based on residuals from a regression of the number of schools built (per district) on the number of school-aged children
- Difference-in-differences estimate of program impact compares pre vs. post differences in treatment vs. comparison communities

Example: A Natural Experiment in Education

The simplest difference-in-differences estimator is:

$$DD = \bar{Y}_{post}^{treatment} - \bar{Y}_{pre}^{treatment} - \left(\bar{Y}_{post}^{comparison} - \bar{Y}_{pre}^{comparison} \right)$$

Dependent Variable: Years of Schooling

	Many Schools Built	Few Schools Built	Difference
Over 11 in 1974	8.02	9.40	-1.38
Under 7 in 1974	8.49	9.76	-1.27
Difference	0.47	0.36	0.12

Difference-in-differences estimation compares the change in years of schooling (i.e. the pre vs. post estimate) in treatment, control areas

- Program areas increased faster than comparison areas
- Difference is not statistically significant

Example: A Natural Experiment in Education

The simplest difference-in-differences estimator is:

$$DD = \bar{Y}_{post}^{treatment} - \bar{Y}_{pre}^{treatment} - \left(\bar{Y}_{post}^{comparison} - \bar{Y}_{pre}^{comparison} \right)$$

Dependent Variable: Log (Wages)

	Many Schools Built	**Few Schools Built**	**Difference**
Over 11 in 1974	6.87	7.02	-0.15
Under 7 in 1974	6.61	6.73	-0.12
Difference	-0.26	-0.29	0.026

Difference-in-differences estimation compares the change in the log of adult wages (i.e. the pre vs. post estimate) in treatment, control areas

- Program had a modest impact on adult wages
- Difference is not statistically significant

DD in a Regression Framework

The DD estimator does precisely that

$$y_{it} = \beta_0 + \beta_1 I\left(treat_{it}\right) + \beta_2 I\left(post_{it}\right) + \beta_3 I\left(treat_{it}\right) \times I\left(post_{it}\right) + \varepsilon_{it}$$

- We need a full panel of firms consisting of Delaware ($I(treat) = 1$) and Non-Delaware ($I(treat) = 0$) registered firms observed before ($I(post) = 0$) and after ($I(post) = 1$) the passage of the law
- β_3 is the parameter of interest (i.e., the DD estimator)

Extension 1: Continuous Treatment

When treatment intensity is a continuous variable:

$$Y_{i,t} = \alpha + \beta Intensity_i + \zeta Post_t + \delta \left(Intensity_i * Post_t \right) + \varepsilon_{i,t}$$

- Instead of using I(treat) dummy, we now consider treatment as a continuous variable
- The effect is then the marginal effect of treatment indensity
- Everything else are the same

Extension 1: Continuous Treatment

Main empirical specification in Duflo (2001):

$$S_{ijk} = \alpha + \eta_j + \beta_k + \gamma\,(\mathit{Intensity}_j * \mathit{Young}_i) + \mathbf{C}_j\delta + \varepsilon_{ijk}$$

where:

- S_{ijk} = education of individual i born in region j in year k
- η_j = region of birth fixed effect
- β_k = year of birth fixed effect
- Young_i = dummy for being 6 or younger in 1974 (treatment group)
- $\mathit{Intensity}_j$ = INPRES schools per thousand school-aged children
- $\mathbf{C}_j$ = a vector of region-specific controls (that change over time)

Extension 1: Continuous Treatment

Dependent Variable: Years of Education

	Obs.	OLS (1)	OLS (2)	OLS (3)
Panel A: Entire Sample				
$Intensity_j * Young_i$	78,470	0.124	0.150	0.188
		(0.025)	(0.026)	(0.029)
Panel B: Sample of Wage Earners				
$Intensity_j * Young_i$	31,061	0.196	0.199	0.259
		(0.042)	(0.043)	(0.050)
Controls Included:				
YOB∗enrollment rate in 1971		No	Yes	Yes
YOB∗other INPRES programs		No	No	Yes

Sample includes individuals aged 2 to 6 or 12 to 17 in 1974. All Specifications include region of birth dummies, year of birth dummies, and interactions between the year of birth dummis and the number of children in the region of birth (in 1971). Standard errors are in parentheses.

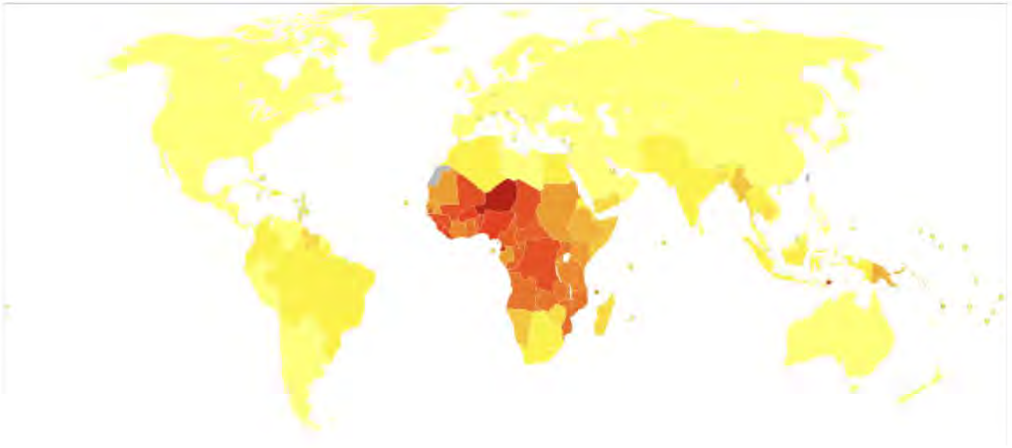
Extension 1: Continuous Treatment

Dependent Variable: Log Hourly Wages (as Adults)

	Obs.	OLS (1)	OLS (2)	OLS (3)
Panel A: Sample of Wage Earners				
$Intensity_j * Young_i$	31,061	0.0147	0.0172	0.027
		(0.007)	(0.007)	(0.008)
Controls Included:				
YOB∗enrollment rate in 1971		No	Yes	Yes
YOB∗other INPRES programs		No	No	Yes

Sample includes individuals aged 2 to 6 or 12 to 17 in 1974. All Specifications include region of birth dummies, year of birth dummies, and interactions between the year of birth dummis and the number of children in the region of birth (in 1971). Standard errors are in parentheses.

Example: Malaria Eradication as a Natural Experiment



Malaria kills about 800,000 people per year

- Most are African children
- Repeated bouts of malaria may also reduce overall child health
- Countries with malaria are substantially poorer than other countries, but it is not clear whether malaria is the cause or the effect

Example: Malaria Eradication as a Natural Experiment





Organized efforts to eradicate malaria are a natural experiment

- First the US (1920s) and then many Latin American countries (1950s) launched major (and successful) eradication campaigns
- Compare trends in adult income by birth cohort in regions which did, did not see major reductions in malaria because of campaigns

Example: Malaria Eradication as a Natural Experiment

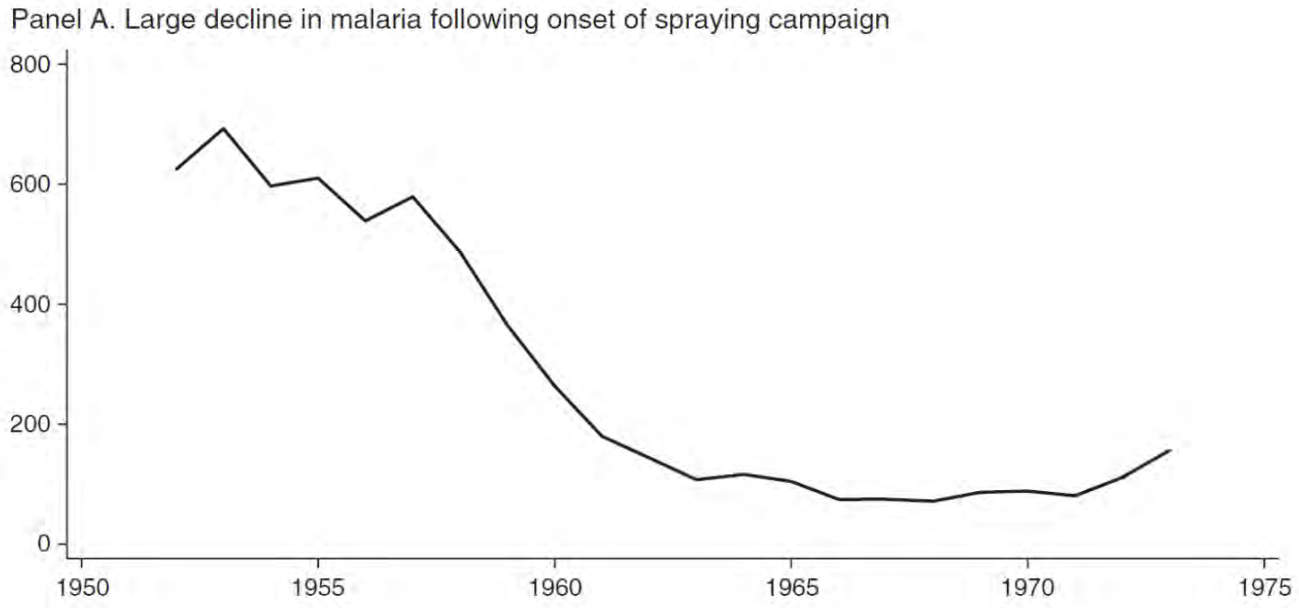
Malaria Eradication in the Americas: A Retrospective Analysis of Childhood Exposure†

By HOYT BLEAKLEY*

This study uses the malaria-eradication campaigns in the United States (circa 1920) and in Brazil, Colombia, and Mexico (circa 1955) to measure how much childhood exposure to malaria depresses labor productivity. The campaigns began because of advances in health technology, which mitigates concerns about reverse causality. Malarious areas saw large drops in the disease thereafter. Relative to non-malarious areas, cohorts born after eradication had higher income as adults than the preceding generation. These cross-cohort changes coincided with childhood exposure to the campaigns rather than to pre-existing trends. Estimates suggest a substantial, though not predominant, role for malaria in explaining cross-region differences in income. (*JEL* I12, I18, J13, O15)

Example: Malaria Eradication as a Natural Experiment

Colombia's malaria eradication campaign began in in the late 1950s...



... and led to a huge decline in malaria morbidity

Example: Malaria Eradication as a Natural Experiment

Areas with highest pre-program prevalence saw largest declines in malaria

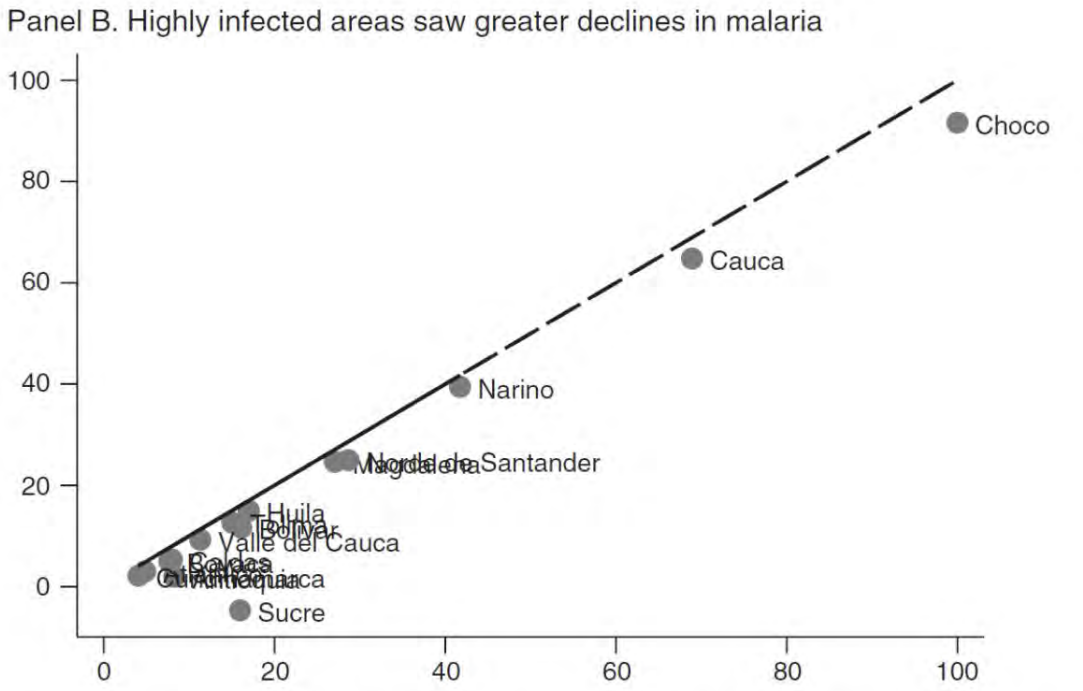


FIGURE 1. MALARIA INCIDENCE BEFORE AND AFTER THE ERADICATION CAMPAIGN, COLOMBIA

Example: Malaria Eradication as a Natural Experiment

In this framework, treatment is a continuous variable

- Areas with higher pre-intervention malaria prevalence were, in essence "treated" more intensely by the eradication program
- Malaria-free areas should not benefit from eradication
- They can be used (implicitly) to measure the time trend

Exposure (during childhood) also depends on one's year of birth

- Colombians born after 1957 were fully exposed to program
  - Did not suffer from chronic malaria in their early childhood
  - Did not miss school because of malaria
- Colombians born before 1940 were adults by the time the eradication campaign began, serve as the comparison group

Example: Malaria Eradication as a Natural Experiment

Regression specification:

$$Y_{j,post} - Y_{j,pre} = \alpha + \beta M_{j,pre} + \delta X_{j,pre} + \varepsilon_j$$

where

- $Y_{j,t}$ is an outcome of interest (eg literacy)
- $M_{j,pre}$ is pre-eradication malaria prevalence
- $X_{j,pre}$ is a vector of region-level controls
- ε_j is the noise term

Example: Malaria Eradication as a Natural Experiment

Regression specification:

$$Y_{j,post} - Y_{j,pre} = \alpha + \beta M_{j,pre} + \delta X_{j,pre} + \varepsilon_j$$

	Malaria ecology (Poveda)			Malaria ecology (Mellinger)		
Dependent variables: Differences across cohorts in…	Literacy	Years of schooling	Income index	Literacy	Years of schooling	Income index
Panel A. Alternative controls						
Additional controls:						
None (basic specification)	0.035***	0.168*	0.065***	0.071***	0.064	0.048***
	(0.013)	(0.088)	(0.011)	(0.016)	(0.108)	(0.014)
Conflict	0.032***	0.175*	0.063***	0.068***	0.068	0.046***
	(0.012)	(0.090)	(0.011)	(0.016)	(0.110)	(0.014)
Economic activity	0.008	0.194**	0.057***	0.043***	0.156	0.039***
	(0.010)	(0.089)	(0.012)	(0.013)	(0.110)	(0.014)
Other diseases	0.024*	0.180**	0.065***	0.058***	0.057	0.042***
	(0.013)	(0.089)	(0.012)	(0.016)	(0.114)	(0.015)
Full controls	0.006	0.165*	0.064***	0.046***	0.076	0.034**
	(0.011)	(0.095)	(0.013)	(0.015)	(0.117)	(0.015)

Extension 2: Event study

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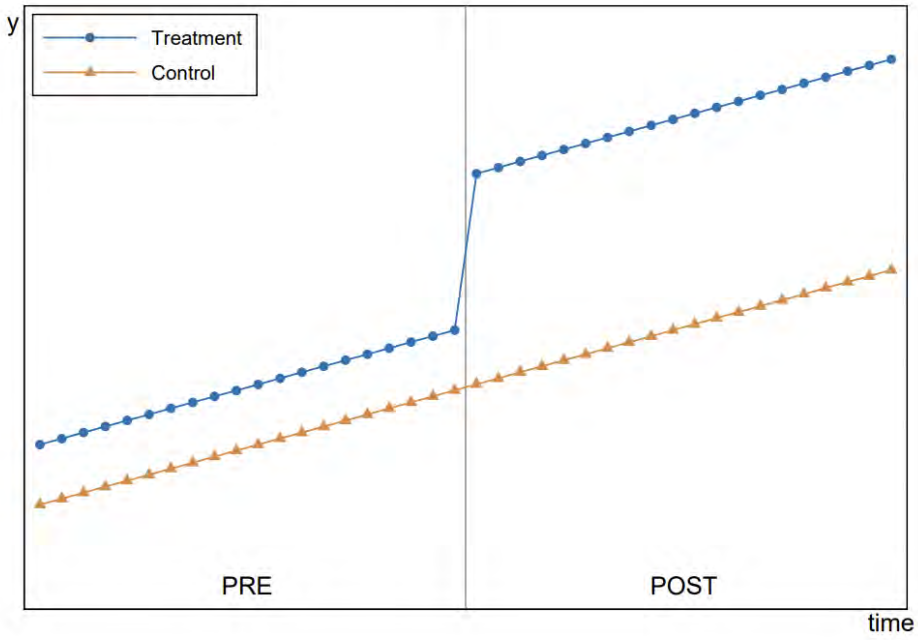
We can also implement diff-in-diff in a panel data framework when more than two periods of data are available; this can increase statistical power*

$$Y_{i,t} = \alpha + \eta_i + \nu_t + \gamma D_{i,t} + \varepsilon_{i,t}$$

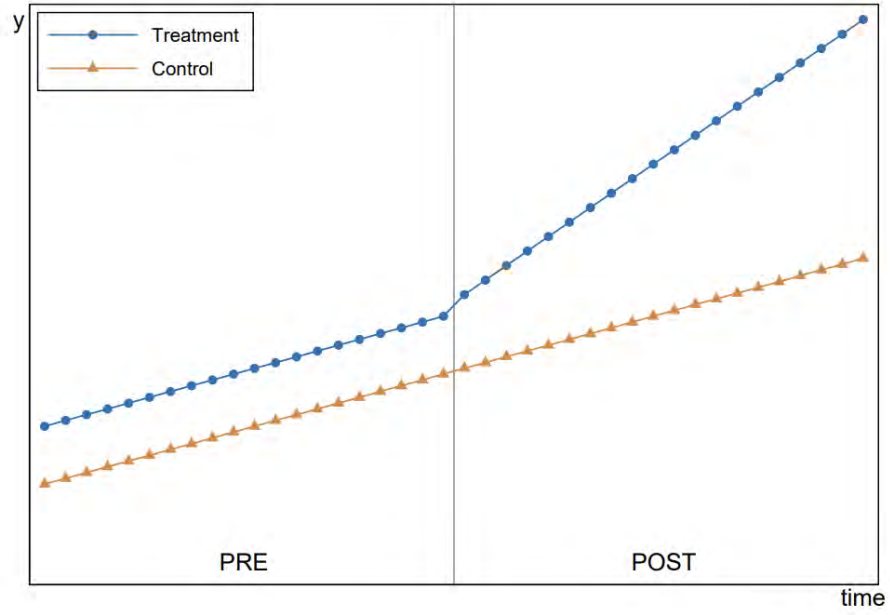
with some caveats:

- Variation in treatment timing?
- Allows for a credible defense of the common trends assumption
  - Unless the common trends assumption is violated
- Serial correlation in treatment and outcome variable is a problem

Extension 2: Event study



Extension 2: Event study



Extension 2: Event study

Event study framework includes dummies for each post-treatment period:

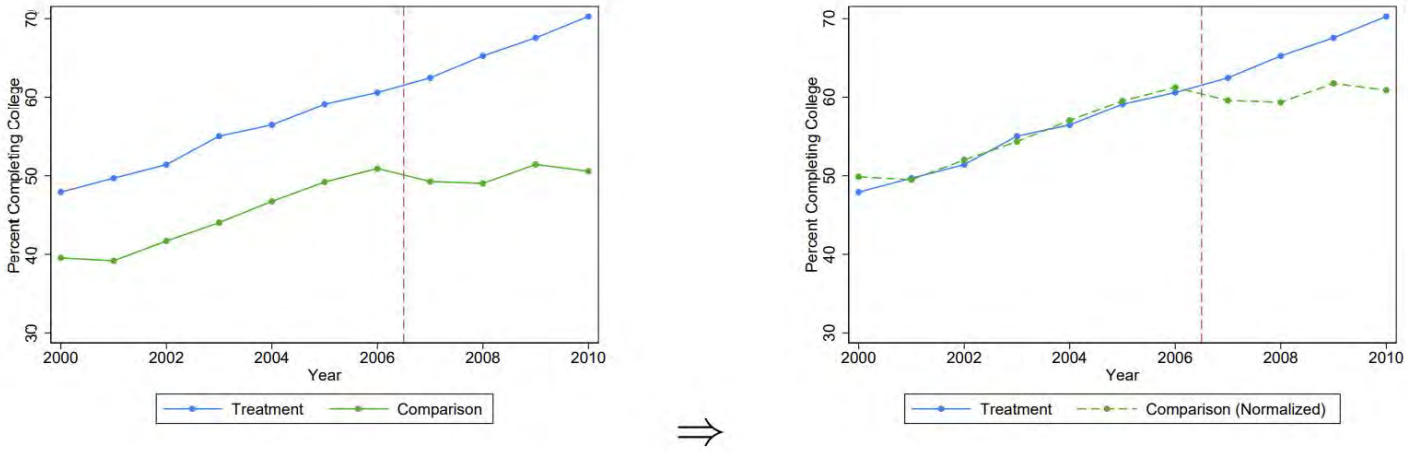
$$Y_{i,t} = \alpha + \eta_i + \nu_t + \gamma_1 D1_{i,t} + \gamma_2 D2_{i,t} + \gamma_3 D3_{i,t} + \ldots + \varepsilon_{i,t}$$

- This new regression allows us to estimate the "dynamic" treatment effect
- Parallel trends assumption for DD is more likely to hold over a shorter time-window
- In the long-run, many other things may happen that could confound the effect of the treatment
- Should be cautious to extrapolate short-term effects to long-term effect

Key Assumption Behind DD

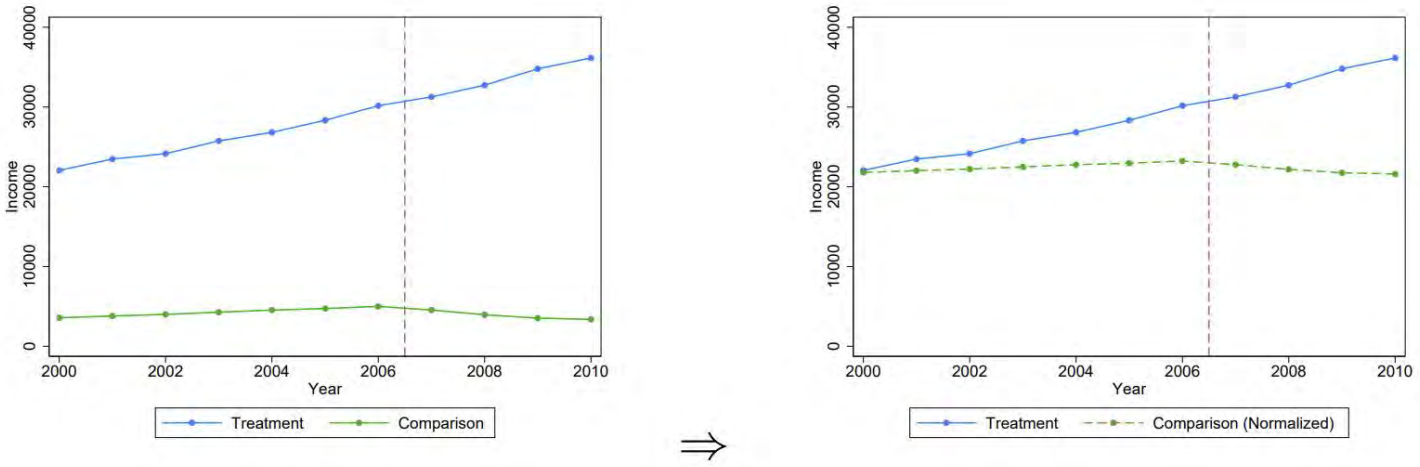
- In the absence of treatment, the average change in the response variable would have been the same for both the treatment and control groups.
  - a.k.a. the **parallel trends** assumption since it requires that the trend in the outcome variable for both treatment and control groups during the pre-treatment era are similar.
- What this assumptions does *not* mean
  - It does *not* mean that the there is *no* trend in the outcome variable during the pre-treatment era (just the same trend across groups)
  - It does *not* require that the *level* of the outcome variable for the two groups be same in the pre-treatment era
    - This can create other issues concerning function form…more on this later

The Common Trends Assumption



Sometimes, the common trends assumption is clearly OK

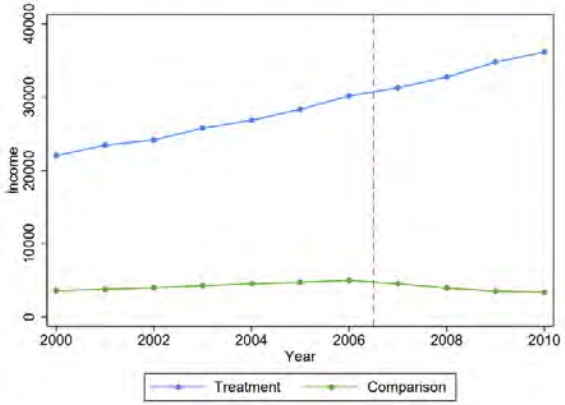
The Common Trends Assumption



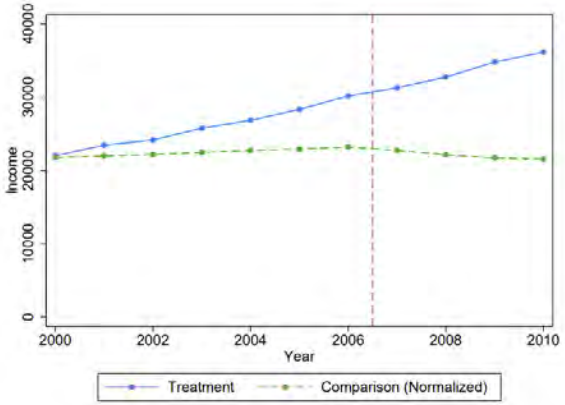
Other times, the common trends assumption is fairly clearly violated

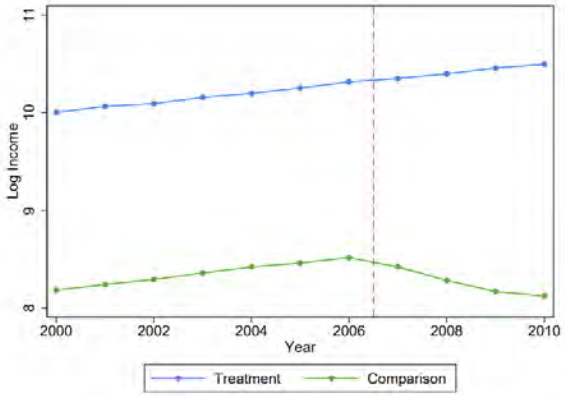
The Common Trends Assumption

Or is it? DD is robust to transformations of the outcome variable

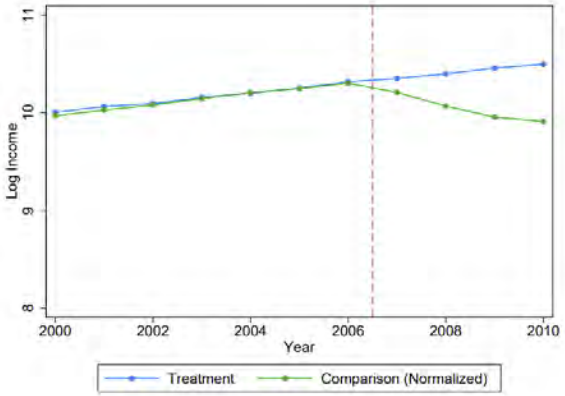


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Defending the Common Trends Assumption

Three approaches:

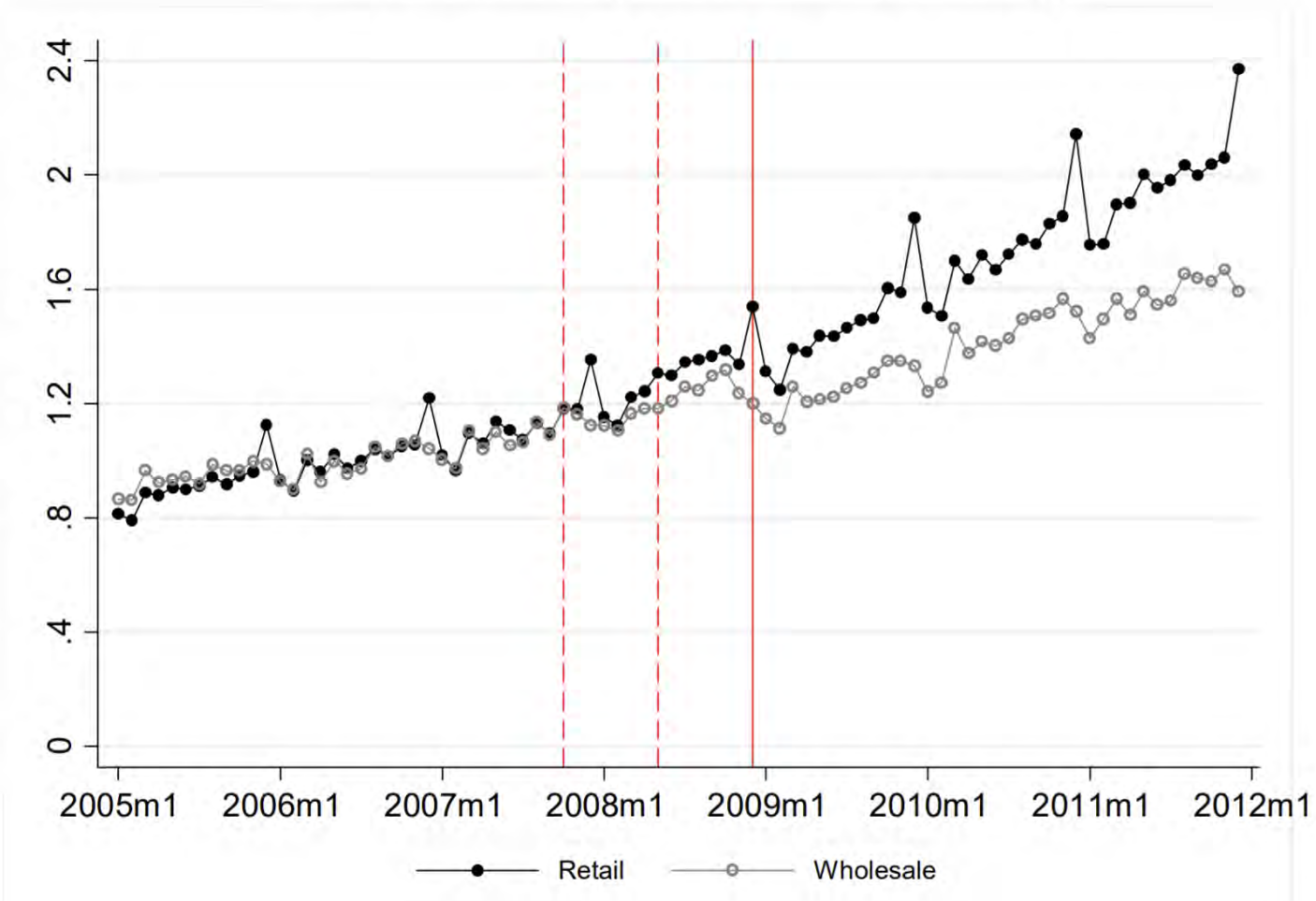
1. A compelling graph
2. A falsification test or, analogously, a direct test in panel data
3. Controlling for time trends directly
    - Drawback: identification comes from functional form assumption

None of these approaches are possible with two periods of data

Approach 1: A compelling graph

Figure 4: Compliance Effect – Retail vs. Wholesale

a. Raw data: reported revenue changes



Approach 1: A compelling graph

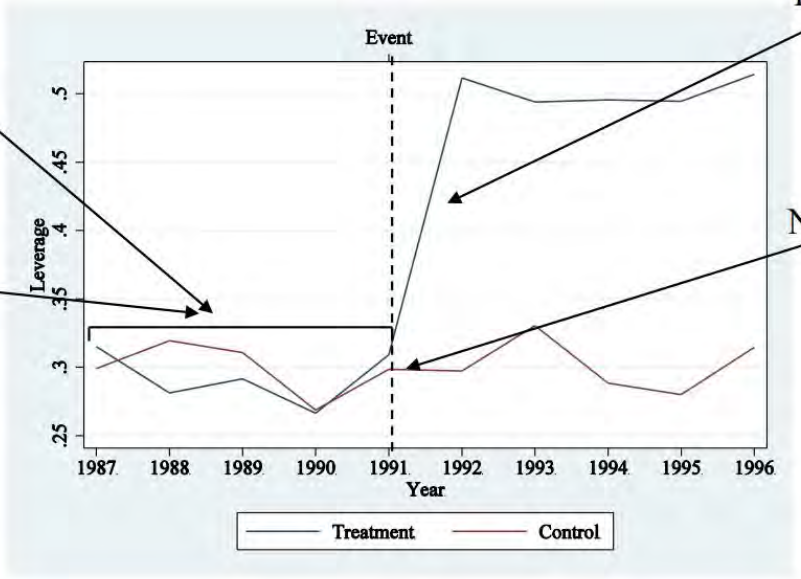
- A "nice" DD picture of our bankruptcy law experiment

Similar leverage growth (~0) in pre-treatment era

Bonus: Similar leverage level (~30%) in pre-treatment era

Sharp change in leverage for treatment group *right after* event

No *change* in leverage after event for control group



Approach 2: Falsification Test

- We can include dummies for each time period

$$Y_{i,t} = \alpha + \eta_i + \nu_t + \gamma_1 D1_{i,t} + \gamma_2 D2_{i,t} + \gamma_3 D3_{i,t} + \ldots + \varepsilon_{i,t}$$

- Ideally, the “treatment effect” should be statistically insignificant before the treatment